Compression as Cartography: HOPE, Tested Against Language

Results from a verified NumPy implementation of Mobahi and Bartlett's "Hilbert Operator for Progressive Encoding" (arXiv:2607.21366), put on trial against a character-level language model, three corpora of increasing breadth, and GPT-2.

When I first read HOPE, I closed my response with a question the paper left open: whether the Gaussian surrogate survives the Zipfian skew of text activations. This is the answer, or at least the beginning of one. The short version: the surrogate dies in correlation everywhere, survives in rank wherever its shape assumption holds, and in the one place that assumption collapses outright, the rank itself inverts. The ordering it assigns to neurons is durable across corpora and most of GPT-2's depth; the redundancy it perceives between neurons never once changes a decision; and in GPT-2's most heavily non-Gaussian block, the data-free map is not merely noisy but anti-guidance. The map, drawn anyway, turns out to be real, robust, and legible; and its legibility is register-relative in exactly the way the paper's transfer method presupposes.

The instrument

The claim under test is that a neuron's identity is the function it computes over a distribution of inputs, not the parameters that happen to encode it. Formally, neuron i with input weights w_i, bias, and output weights w_i^{out} is lifted to a rank-1 operator whose norm, its **capacity**, is

$$\|f_i\| \;=\; \|w_i^{out}\|\sqrt{K_{ii}}, \qquad K_{ii} \;=\; \mathbb{E}\big[\phi(y_i)^2\big]\,,\; y_i \sim \mathcal{N}(\beta_i, \gamma_i^2).$$

Everything downstream depends on evaluating such kernels without data. The surrogate declares each pre-activation Gaussian with calibration-measured mean and standard deviation; for ReLU the self-kernel is closed-form (the paper's Eq. 3),

$$K_{ii} \;=\; (\beta^2+\gamma^2)\,\Phi(\beta/\gamma) \;+\; \beta\gamma\,\varphi(\beta/\gamma),$$

pairwise correlation comes from weight geometry through the Maximum-Entropy warp (Eq. 4), solving $\hat\rho/(1-\hat\rho^2) = \kappa$ with

$$\kappa \;=\; \frac{\rho_{\text{eff}}}{1-\rho_{\text{eff}}^2}\, r_i r_j, \qquad r_k = \sigma_k/\|w_k\|, \qquad \hat\rho = \frac{2\kappa}{1+\sqrt{1+4\kappa^2}},$$

and the cross-kernel follows the arc-cosine form (Eq. 5), $K_{ij} \approx J_1(\hat\rho)\sqrt{K_{ii}K_{jj}}$ with $J_1(\rho) = \left(\sqrt{1-\rho^2} + (\pi - \arccos\rho)\,\rho\right)/\pi$. Pruning and merging are then priced in one currency (Eq. 6): with layer capacity E_a,

$$J_{\text{prune}}(i) = \frac{N\|f_i\|}{E_a - \|f_i\|}, \qquad J_{\text{merge}}(i,j) = \frac{N\cdot D}{E_{\text{rem}} + s^*},$$

where the optimal parent magnitude and distortion are $s^* = (a + b\,E_{\text{rem}})/(2E_{\text{rem}}+b)$ and $D^2 = a + 2{s^*}^2 - 2s^*b$, with a the pair's Hilbert energy, b its alignment with the synthesized parent direction (the principal right-singular vector of $w_i^{out}\tilde w_i^\top + w_j^{out}\tilde w_j^\top$, computed by the rank-2 reduction), and E_{rem} the capacity of everything else. Every closed form in the implementation is pinned by Monte-Carlo tests; for

GPT-2, whose GELU sits outside the PH-1 family the closed forms cover, the same kernels are evaluated by Gauss-Hermite quadrature under the same surrogate, pinned both against Monte Carlo and against the ReLU closed forms.

The experimental design was staged: build and verify the engine; put the surrogate on trial (are pre-activations Gaussian? do the kernels track reality?); then draw the map, deploying every action into the live model so the ledger doubles as a loss trajectory, with each unit carrying a stable identity, a linguistic label, and its measured kurtosis. Each mapping run was performed twice, once with the paper's warped correlations and once with correlations measured on the calibration slice, to isolate how much of the map depends on the correlation model. The corpora: tiny Shakespeare (1.1M characters), the MASC slice of the Open American National Corpus (2.9M characters, 20 genres), and the full OANC (88.5M characters, 8 genres), with GPT-2 small as the transformer frontier.

Finding 1: the fiction holds in the middle and fails in the tail

On every model tested, the median unit is close to Gaussian and the tail is not. The char-MLP's median excess kurtosis is -0.008 (max $+3.1$); MASC gives -0.030 (max $+7.3$); full OANC $+0.008$ (max $+7.3$). GPT-2 is the same story amplified: pooled median $+0.43$, maximum $+776.6$, with the non-Gaussianity concentrated at the ends of the stack (block 2's median excess kurtosis is $+2.4$; the mid-stack plateaus near $+0.22$; block 12 rises again). The Zipfian skew of text does reach the activations, but it arrives as a heavy tail of outlier units, not as a failure of the typical case. Kernel values feel it: self-kernel errors run 7 to 17 percent across the corpora and cross-kernels 9 to 25 percent, growing with breadth.

Finding 2: distances distort; the ordering survives

The number the framework actually spends is not the kernel value but the ranking it induces. Capacity orderings correlate with empirically measured capacities at Spearman 0.982 (Shakespeare), 0.963 (MASC), and 0.938 (OANC): a visible decay with distributional breadth, entangled with the models being better trained, but an ordering that remains firmly intact. A stranger regularity replicated three times: the full data-free pipeline predicts cross-kernels *better* than the same pipeline given oracle correlations, because the MaxEnt warp's shrinkage partially cancels the zero-bias approximation's overshoot. Two fictions, leaning on each other.

Finding 3: the warp silences consolidation

The headline. Under the paper's correlation model the encoder **never merges**. Across all three corpora, every single action chosen was a prune: 167 of 167, 250 of 250, 249 of 249. The mechanism is visible in one pair of numbers: on the char-MLP the warp caps pairwise correlation at 0.13 where the empirical maximum is 0.67. No pair ever looks redundant enough for a merge to outprice a prune. Replace $\hat{\rho}$ with measured correlations, leaving every other step untouched, and merging wakes: 17, 41, and 76 merges respectively, with slightly better loss trajectories. The framework's elegant unification of pruning and merging into one projection is, on text-trained models under its own surrogate, a unification in principle only; removal is priced truthfully, however redundancy is not perceived.

Finding 4: the map is real, robust, and corpus-relative

Judged as compression, the capacity ordering earns its keep where there is slack to find: HOPE's trajectory sits 0.47 to 0.55 nats below L1-norm pruning at 35 percent density on the smaller corpora, and

0.43 below on full OANC. At full scale a caveat appears that I consider a finding: random pruning ties capacity-ordered pruning (2.638 vs 2.636 nats), because a saturated model has no cheap units, so every order costs alike. The value of the ordering then concentrates entirely in what survives rather than the curve, which is the cartographic point anyway.

Judged as measurement, the map barely moves when the correlation model is swapped: removed-set Jaccard 0.898, 0.912, 0.789 and removal-order Spearman 0.974, 0.946, 0.899 across the three corpora. And what the map says is legible and corpus-relative in the way a map should be. On Shakespeare, structure survives: the newline and speaker-header detectors (one unit's top-activating context is literally `\nWARWICK`) are among the best preserved, while punctuation detectors are cut hardest. On OANC the same architecture, allocated against different territory, evicts its newline units entirely; the corpus defines what counts as load-bearing. Heavy-tailed units skew toward surviving in every run, consistent with outlier features being high-capacity detectors of frequent structure: the transformer outlier-feature worry attaches to mispricing, not to being pruned first.

Finding 5: the register-conditional map, or DEFT's premise measured

Because the surrogate is defined by whatever distribution calibration sees, the same trained weights can be mapped per register. On OANC's genres this yields a mostly shared core (median cross-register capacity Spearman 0.979, top-half core overlap 0.90) with a measurable, legible fringe: the poles are conversation and technical prose (minimum Spearman 0.855, core overlap 0.745), and the most register-variable units read exactly as they should, transcript-formatting detectors core in conversation, biomedical text detectors core in technical writing, an all-caps onset detector core in movie scripts and slack in travel guides. Which knowledge is core is, measurably, a property of the register the surrogate protects. That is the premise DEFT's frozen-core transfer stands on, demonstrated on real registers with nameable units.

Finding 6: GPT-2, where the fiction's failure reaches the map

The GPT-2 experiment (blocks 2, 6, and 12 encoded to 35 percent density, OANC calibration and evaluation, quadrature GELU kernels) was designed around one hypothesis: paper-versus-empirical map divergence should track the depth profile of non-Gaussianity. The hypothesis resolved through a different channel than posed, twice over.

First, correlations never mattered. Zero merges under both correlation models in every block; removed-set Jaccard 1.000 three times; the paper, empirical, and capacity-prune-only trajectories are pixel-identical. The mechanism is width: a 3072-unit block carries so deep a reservoir of near-dead units that the cheapest prune undercuts every candidate merge's distortion all the way down to 35 percent density. On the char-MLP the warp blinded the encoder to redundancy; on GPT-2, consolidation is outcompeted even with redundancy in plain view. The correlation half of the framework is moot at transformer width, at least at these densities.

Second, the fiction's failure surfaced in the capacity pricing itself, and it surfaced exactly where the shape statistics said to look. In block 2, whose median excess kurtosis is +3.09 under OANC calibration, the data-free ordering is anti-guidance: following it drives NLL from 3.56 to roughly 5.15 by 60 percent density, then recovers non-monotonically to 4.3, while random deletion of the same fractions costs almost nothing. Two things in that sentence deserve emphasis. Random being free means block 2 is highly redundant to unbiased deletion, so the surrogate is not failing to find slack; it is systematically selecting load-bearing units as slack. And the non-monotone recovery means the damage involves interactions between units that no per-unit pricing can represent. Block 6's ordering trails random mildly; block 12's beats every baseline. Three blocks are three data points, so the honest claim is locational rather than dose-response: the catastrophic inversion co-locates with the extreme non-Gaussianity, and where the Gaussian fiction roughly holds, the ordering is competitive to winning.

The register-conditional map, meanwhile, gained a depth axis, and it is the run's most consequential result: cross-register capacity Spearman declines near-monotonically from about 0.90 in blocks 1 through 3 to 0.48-0.55 in blocks 10 through 12 (with a small end-of-stack uptick). Register-universal foundations; register-specific depths. The profile is decoupled from the kurtosis curve, which spikes early while register-specificity accumulates late, so the depth gradient is not a heavy-tail artifact. For DEFT this turns "freeze a core" into a measured, depth-resolved partition: for register adaptation, the deep feed-forward blocks are what you would open. For anyone calibrating a data-free map, it yields a rule of practice: the choice of calibration corpus barely matters early in the stack and matters enormously by block 10.

What it means, and what it doesn't

For HOPE: the correlation model fails conservatively (missing merges rather than making bad ones) and, at transformer width, fails into irrelevance; the capacity model is the load-bearing joint, trustworthy across corpora and across most of GPT-2's depth, and anti-trustworthy precisely where the shape assumption collapses. The practical diagnostic is already in hand: the same calibration pass that builds the surrogate yields the per-block kurtosis profile that flags where not to believe it. For language models: compression order is readable as a hierarchy of dispensability, that hierarchy is corpus- and register-relative, and the eviction ledger gives the "acquisition in reverse" reading an operational form. The limits being these are small models mapped without retraining between actions (deliberately, to read the raw data-free trajectory); labels are cheap correlational probes; the breadth trend is confounded with training quality; and the GELU parent synthesis fixes an input scale that ReLU's homogeneity would have left free.

Engineering notes

The stack was chosen for verifiability. The engine and both pilot models are pure NumPy and SciPy: closed-form kernels, merge algebra, and the character MLP itself. The point of refusing a framework was to leave no distance between the equations in this document and the code that executes them. PyTorch and Hugging Face transformers appear only where GPT-2 itself lives; even there, the analysis stays in NumPy, with weights extracted, priced, and written back as arrays.

Verification is treated as architecture, not afterthought. Every layer of the tower is pinned by the 34-test suite: closed forms against Monte Carlo; the O(n) rank-2 parent solve against an explicit SVD reference; the Gauss-Hermite GELU kernels against both Monte Carlo and the ReLU closed forms they generalize; writeback exactness meaning a pruned model must equal manual subtraction to 1e-10, a merged parent must realize its predicted function; the vectorized correlation warp against its scalar reference; and the incremental pairwise tracker against a from scratch rebuild to 1e-9. The GPT-2 script additionally opens with runtime self-checks like hook capture semantics and writeback equivalence designed to fail before compute is spent.

Everything is resumable, cached, and deterministic. Each driver decomposes into idempotent parts - train, surrogate, map, registers, finish - with artifacts cached between them: model weights as npz, cleaned corpora, BPE token streams, pickled map states. Seeds are fixed throughout, smoke tests write to an isolated results directory, and any part can be rerun in seconds from its caches.

Two optimizations made the mapping loops feasible. Replacing the per-pair SVD in parent synthesis with the equivalent 2x2 eigenproblem took a full mapping run from exceeding its compute ceiling to twelve seconds. At GPT-2 width, maintaining the pairwise structure incrementally (one Gram or correlation build, then rank-local row and column updates as units leave and parents arrive) replaced a quadratic per-step rebuild that would have turned each block's encoding into hours. The remaining costs are explicit knobs: a top-correlated candidate prefilter for merge pricing, and loss checkpoints every 64 actions rather than every action.

The data layer is where language projects live or die. The corpus module anchors OANC's genre labels on its directory tiers, detags and detokenizes the MASC slice from an independent mirror, splits train from validation at document level stratified by genreso n o document leaks across the boundary, samples calibration contexts within register spans, and encodes characters as uint8 (85 MB where int64 would have been 680). Acquiring the corpus itself required engineering around a dormant host with an expired certificate including pinned plain-HTTP commands, size and structure checks, and cross validation of the download against the independently mirrored MASC documents it contains.

Writeback maintains the ledger quality. Every prune and merge is deployed into the live model by direct weight surgery, so every loss curve in this document is measured on the edited model rather than predicted by the theory being tested. The repository follows a src-layout package installable with pip install -e ., with the pilot stages frozen as reported and the broad-corpus and GPT-2 drivers parameterized on top of the same core.